Methodology of Measuring Herring Spawn on Vancouver and Hornby Islands

Introduction

- Pacific Herring are a keystone species on the coast of BC (BC Seafood Alliance, n.d.), in addition to being an integral cultural element of coastal Indigenous groups (Strong Coast, n.d.).
- Each spring, Herring return to the coast to breed (BC Seafood Alliance, n.d.). The herring milt makes the water appear brighter than surrounding areas, often appearing milky turquoise and opaque in nature. The characteristic changes in ocean colour serve as a metric with which herring spawn populations can be measured and tracked (Qi et al., 2021).
- Creating a method with which herring spawning patterns can be measured via satellite imagery is important because spawning patterns provide insight into the health of the ecosystem in this region while providing an additional reliable, long term method of quantifying and monitoring herring spawn patterns.

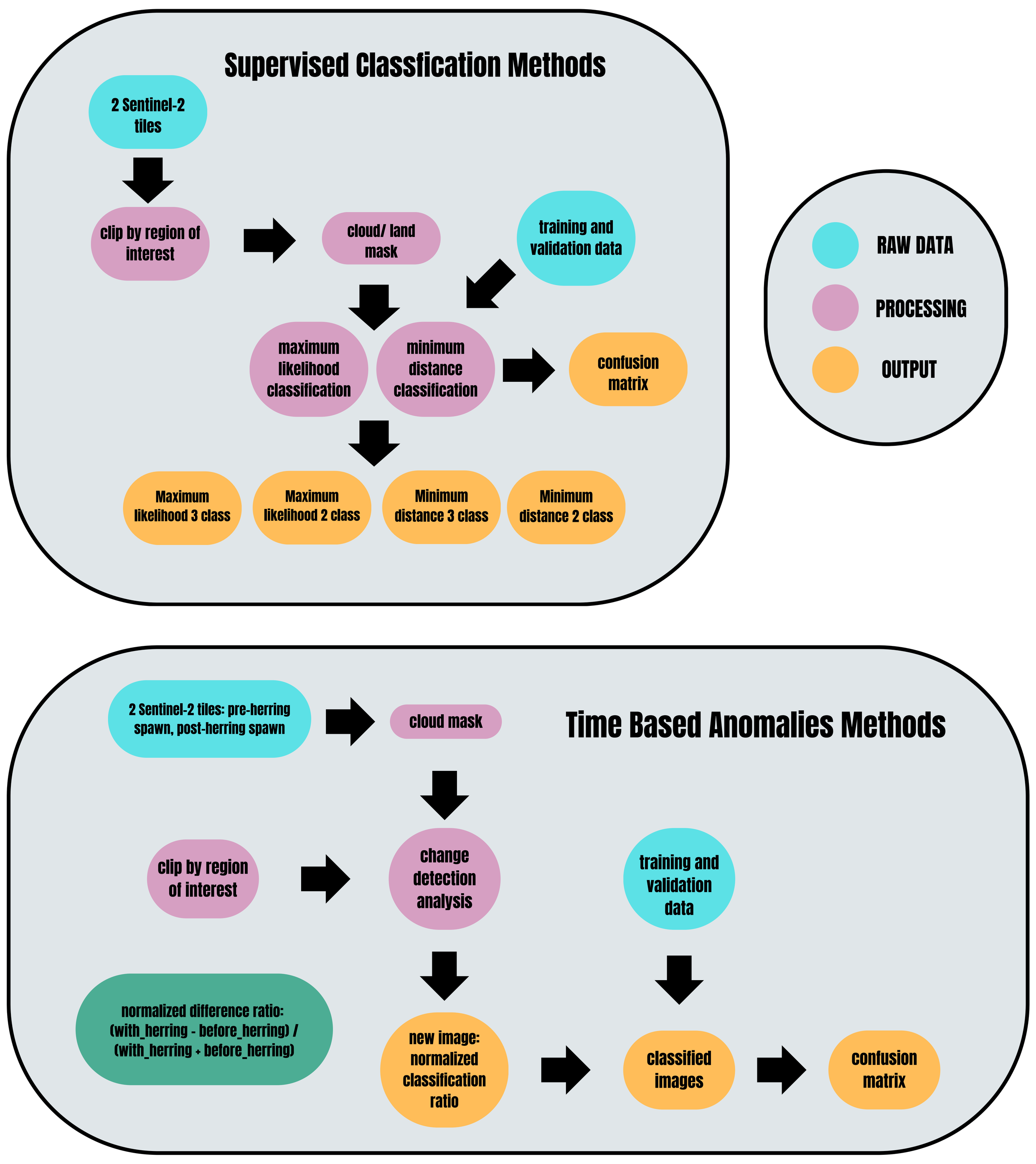
Objective

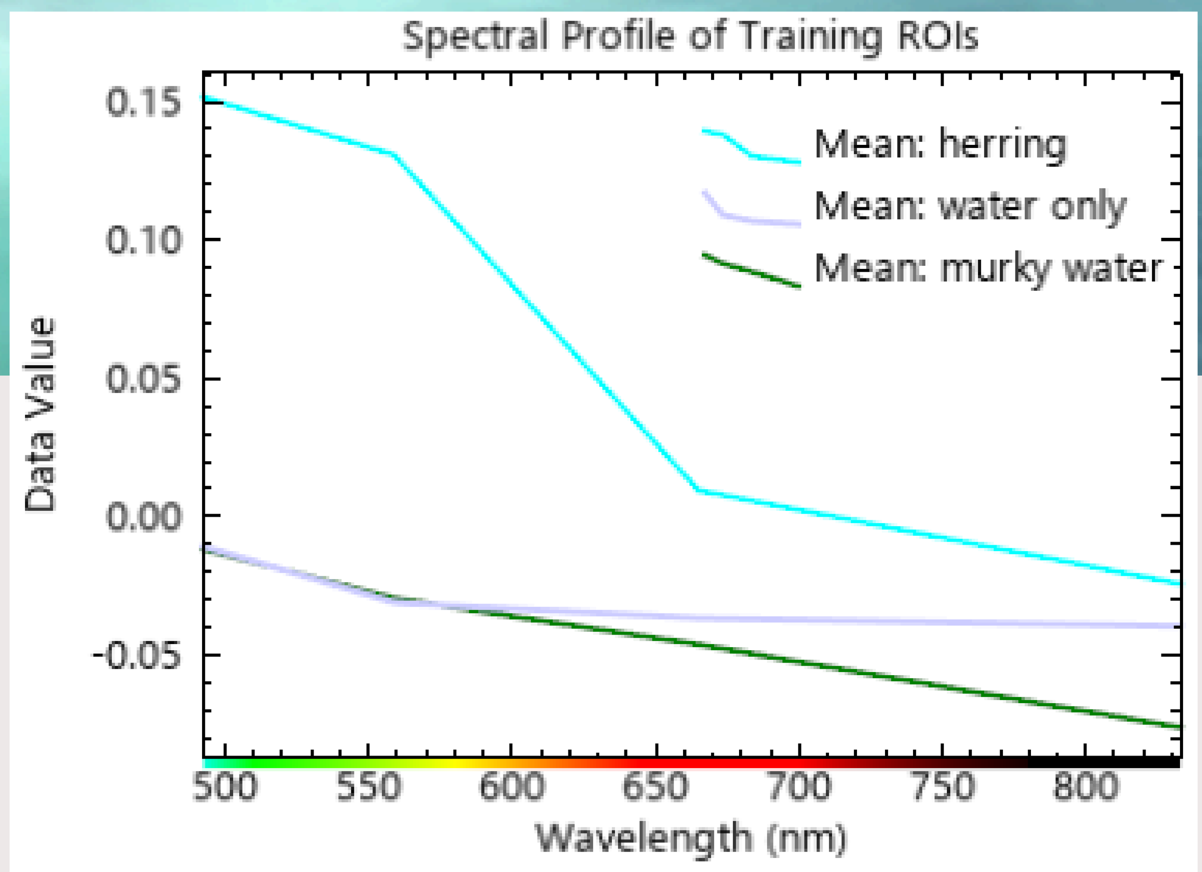
- Determine the efficacy of using remote sensing to classify regions of herring spawn
- To determine the most effective classification method for assessing herring spawn, comparing two methods: supervised learning and time-based anomalies, each with multiple classification methods.
- To demonstrate the characteristics of herring spawn via a spectral index

Data Collection

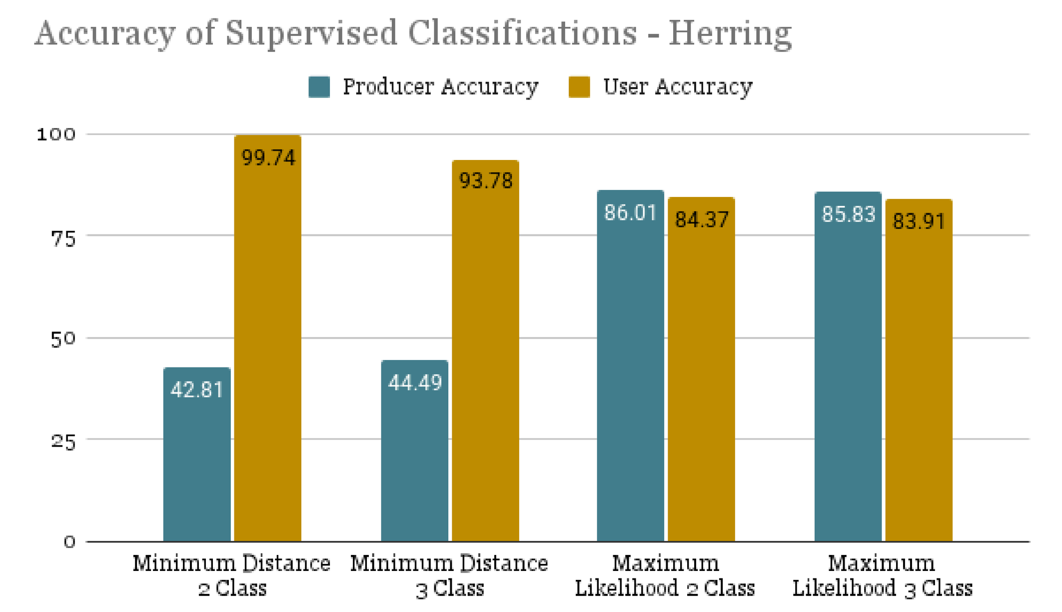
- Sentinel-2 data was chosen based on visibility (cloud cover) and visual intensity of herring spawn, along with its high spatial resolution.
- For images showing herring spawn, availability of validation data was taken into account
- Image with herring spawn: March 9th, 2020
- Clear image without herring spawn: October 26th, 2019 - This image was used for the time-based anomalies method.

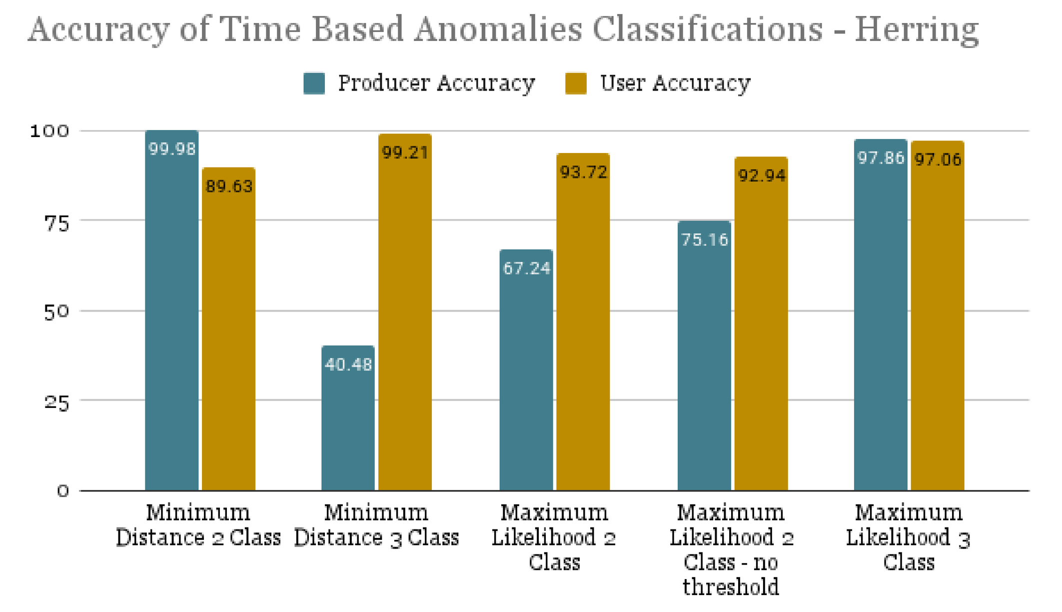
Methods

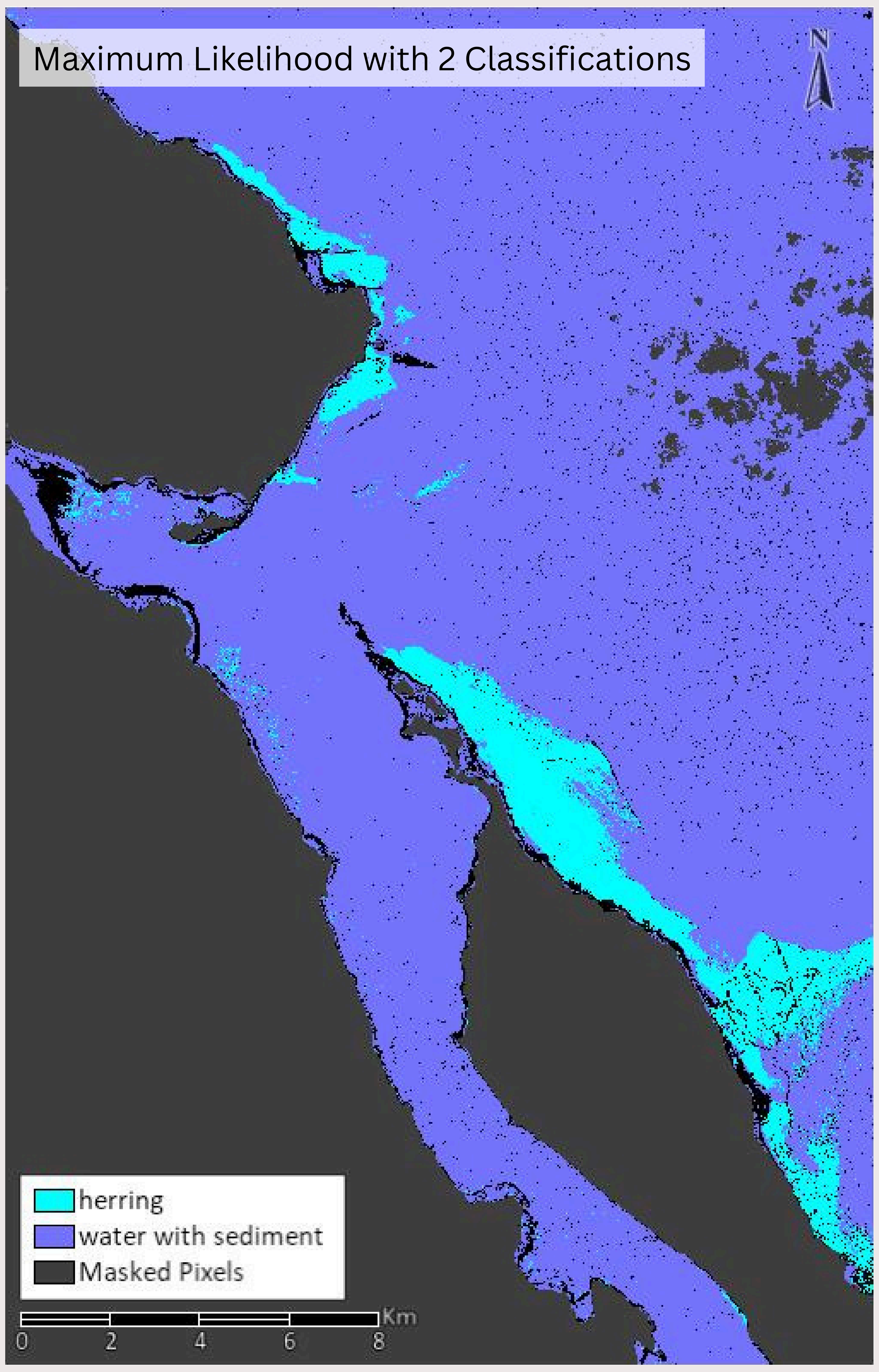


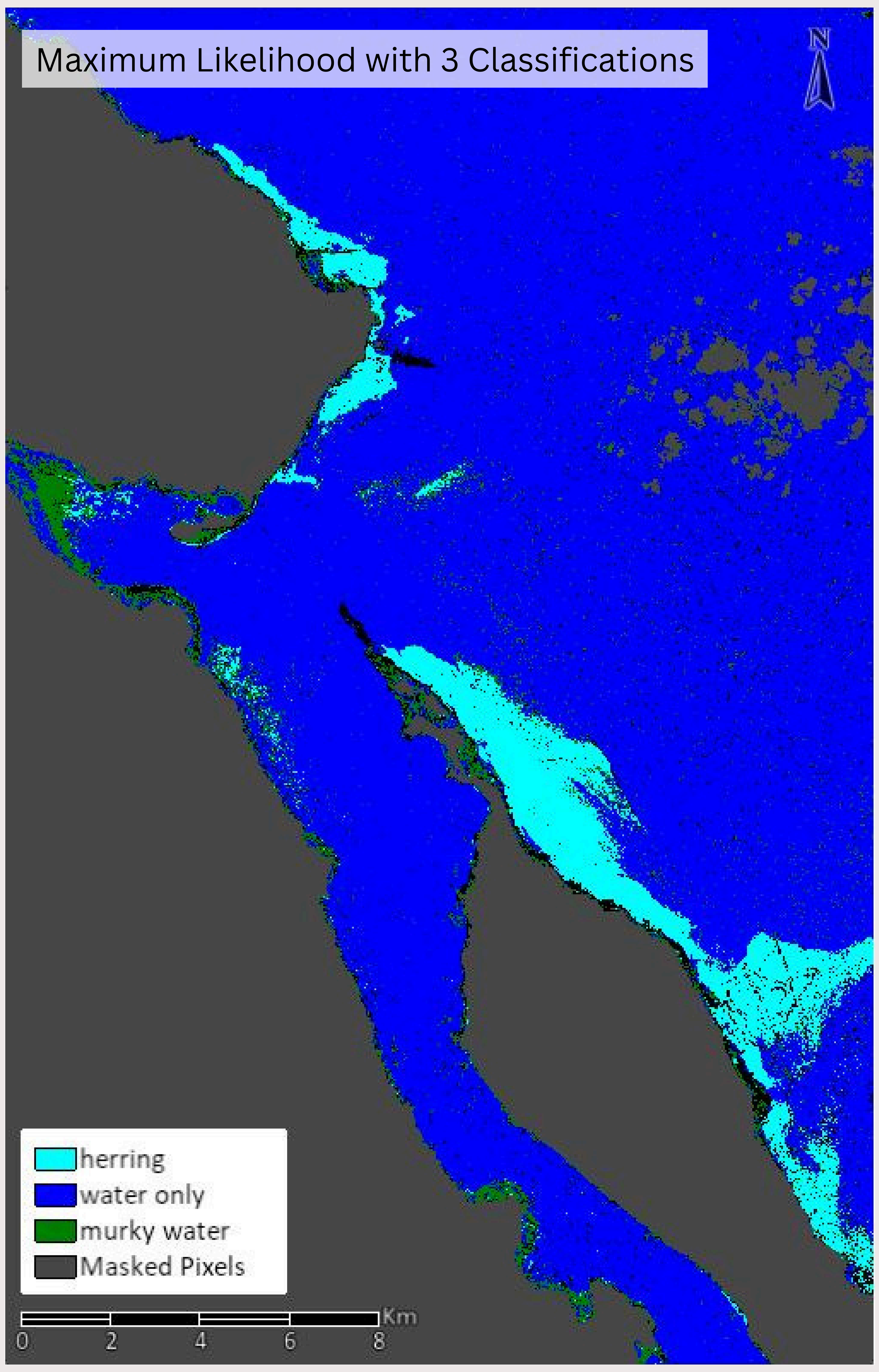


Accuracy Assessment









Results and Discussion

- We found that the time-based anomalies approach generally performed better than the supervised classification approaches that used a single image of the herring spawn. The most consistent accuracy was Maximum Likelihood Classifications within the time-based anomalies method, both 2 and 3 classes. The biggest disparity in Producer and User Accuracy was with Minimum Distance (3-class) Classification. Generally, User Accuracy was higher than Producer Accuracy across Maximum Likelihood classification within the time-based anomalies.
- A limitation of our analysis that we would've liked to explore further but was outside the scope of our project, was the difference in spawning events across a season or years.
- Some sources of possible error or sensitivity with our results relate to the validation data we used. The available ground-truth data did not include precise polygons that we could map to ROIs. This analysis also does not consider the density of spawn in the region. We are unsure how the models' performances differ between high and low-density spawn areas.

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